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Project

Hessian-Based Mixed-Precision Quantization for Vision Transformers

MMML Project Report

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Abstract

This project investigates whether second-order information about the loss landscape can guide mixed-precision quantization in Vision Transformers (ViTs) more reliably than uniform low-bit quantization. Specifically, we study Hessian-based sensitivity estimates with emphasis on the Hessian trace and related curvature-aware criteria to determine which model components should be quantized conservatively and which can tolerate aggressive compression. We implement and evaluate the HAWQ and HAWQ-V2 frameworks, connecting numerical analysis and optimization theory with an applied model-compression workflow. Our central hypothesis is that Hessian-guided bit-width allocation improves the trade-off between memory efficiency and predictive accuracy compared to uniform quantization baselines.

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1. Introduction

1.1. Motivation

Modern AI research increasingly demands running compute-intensive models on hardware with tightly constrained memory resources. For transformer-based architectures, and particularly Vision Transformers (ViTs), the dense matrix multiplications within Multi-Head Self-Attention mechanisms make memory a primary deployment bottleneck. The predominant technique for reducing model footprint is *quantization*: projecting continuous, high-precision weights into a discrete, low-bit integer space.

However, the prevailing industry standard of *uniform quantization* in short applying the same bit-width to every layer, fundamentally ignores the mathematical topology of the loss landscape. Different Attention Heads and linear projections reside in entirely different regions of that landscape. Forcing a uniform quantization grid mathematically applies a perturbation vector $\Delta w = w_q - w$ to every layer identically, where w denotes the optimal continuous weights found by the optimizer and w_q denotes the nearest point on the low-bit integer grid. If a layer acts as a highly sensitive architectural bottleneck, this perturbation causes a sharp spike in the model’s loss, destroying predictive accuracy. If a layer sits in a flat region of the landscape, the same perturbation is largely harmless.

Transitioning from heuristic, trial-and-error calibration to a rigorous, calculus-based decision rule is therefore both practically significant, as it enables the safe deployment of foundation models on resource-constrained hardware, and mathematically interesting, as it requires bringing together Taylor expansion theory, second-order matrix analysis, and stochastic trace estimation.

1.2. Hypotheses

We formulate three testable hypotheses that guide the project:

1. **H1: Sensitivity ordering** The normalized Hessian trace $\frac{1}{n_i} \text{Tr}(H_i)$ of a layer i provides a reliable relative ordering of quantization sensitivity: layers with larger normalized trace sustain greater loss increase under equal quantization perturbation.
2. **H2: Mixed-precision benefit** A mixed-precision allocation derived from Hessian sensitivity scores achieves higher accuracy than uniform quantization at the same average bit-width budget.
3. **H3: Trace vs. eigenvalue** The average Hessian trace (HAWQ-V2) provides a strictly more informative sensitivity criterion than the top Hessian eigenvalue (HAWQ), because quantization noise is distributed across the full eigenspectrum rather than concentrated in a single direction.

1.3. Research Goals

Concretely, this project aims to:

1. Implement Hutchinson’s stochastic algorithm to approximate the Hessian trace $\text{Tr}(H_i)$ layer-wise, isolating individual Attention Heads and MLP blocks.
2. Use the computed sensitivity scores to allocate bit-widths: aggressive compression for low- $\text{Tr}(H)$ layers, conservative precision for high- $\text{Tr}(H)$ layers.
3. Compare the resulting mixed-precision model against a uniform quantization baseline and evaluate whether H1-H3 are supported by empirical evidence.

2. Problem Setting

2.1. Quantization as a Mathematical Operation

We divide the network into b blocks $\{W_1, W_2, \dots, W_b\}$, where block i contains n_i learnable scalar parameters.

Definition 2.1 (Uniform quantizer). *A uniform scalar quantizer with B bits maps a continuous value $z \in \mathbb{R}$ to a discrete codebook of 2^B levels via a step function:*

$$Q_B(z) = q_j \quad \text{for } z \in (t_j, t_{j+1}],$$

where $\{t_j\}$ are uniformly spaced thresholds and $\{q_j\}$ are the corresponding reconstruction levels. Applying Q_B element-wise to W_i induces the quantization perturbation

$$\Delta W_i = Q_B(W_i) - W_i.$$

In mixed-precision quantization, each block i is assigned its own bit-width B_i , which may differ from block to block. The goal is to find the assignment $\{B_i\}$ that minimises loss degradation subject to a hardware memory constraint.

2.2. Second-Order Taylor Approximation of Loss Degradation

Let $L(W)$ denote the task loss. At the point of deployment, a pre-trained model has converged to a local minimum, so the first-order gradient satisfies $\nabla_{W_i} L \approx 0$.

Theorem 2.1 (Quadratic loss approximation). *Under the assumption that the model is at a local minimum and cross-block interactions are negligible (block-diagonal Hessian), the change in loss due to quantization of block i is approximated by*

$$\Delta L_i \approx \frac{1}{2} \Delta W_i^T H_i \Delta W_i, \tag{1}$$

where $H_i = \nabla_{W_i}^2 L$ is the Hessian of the loss with respect to the weights of block i .

Proof. Expand $L(W_i + \Delta W_i)$ to second order:

$$L(W_i + \Delta W_i) \approx L(W_i) + \underbrace{\nabla_{W_i} L^T}_{\approx 0} \Delta W_i + \frac{1}{2} \Delta W_i^T H_i \Delta W_i.$$

Dropping the first-order term and subtracting $L(W_i)$ yields (1). $\square$

Remark 2.1. The block-diagonal assumption zeroes out all cross-derivatives $\frac{\partial^2 L}{\partial W_i \partial W_j} = 0$ for $i \neq j$, so the total loss increase decomposes additively: $\Delta L \approx \sum_i \Delta L_i$.

2.3. Hessian Spectrum and Quantization Sensitivity

Equation (1) links layer sensitivity directly to the curvature of the loss landscape as encoded by H_i . To make this operational, we need a scalar summary of H_i that can be computed efficiently and ranked across layers.

HAWQ: top eigenvalue. The original HAWQ framework approximates $H_i \approx \lambda_{\max}^{(i)} I$ (isotropic approximation), giving

$$\Delta L_i \approx \frac{1}{2} \lambda_{\max}^{(i)} \|\Delta W_i\|_2^2.$$

The sensitivity metric is $S_i = \lambda_{\max}^{(i)} / n_i$.

HAWQ-V2: normalized Hessian trace. Decomposing $\Delta W_i = \sum_j a_j v_j$ in the eigenbasis of H_i gives

$$\Delta L_i = \frac{1}{2} \sum_j \lambda_j^{(i)} a_j^2,$$

where $\lambda_j^{(i)}$ are the eigenvalues and $a_j = v_j^T \Delta W_i$ the projections of the quantization noise onto each eigendirection. Since quantization noise is spread across *all* eigendirections rather than concentrated in the direction of $\lambda_{\max}^{(i)}$, the relevant summary is the full curvature average.

Lemma 2.2 (Trace ordering). *Under the assumption that quantization noise ΔW_i is isotropically distributed (i.e., $\mathbb{E}[a_j^2]$ is equal for all j), the expected loss increase satisfies*

$$\mathbb{E}[\Delta L_i] \propto \frac{1}{n_i} \text{Tr}(H_i) = \frac{1}{n_i} \sum_j \lambda_j^{(i)}.$$

Hence, for equal perturbation norm, the layer with larger normalized trace incurs larger expected loss.

This lemma underpins Hypothesis H3: the normalized trace is a strictly more informative sensitivity criterion than $\lambda_{\max}^{(i)}$ alone, because it aggregates curvature over all eigendirections.

2.4. Hutchinson’s Stochastic Trace Estimator

Explicitly forming $H_i \in \mathbb{R}^{n_i \times n_i}$ is infeasible for any realistic ViT block. Instead, HAWQ-V2 employs a matrix-free, randomized trace estimator.

Theorem 2.3 (Hutchinson’s identity). *Let $z \in \mathbb{R}^{n_i}$ be a random vector with i.i.d. entries satisfying $\mathbb{E}[zz^T] = I$ (e.g., Rademacher or Gaussian). Then*

$$\text{Tr}(H_i) = \mathbb{E}[z^T H_i z].$$

Proof.

$$\mathbb{E}[z^T H_i z] = \mathbb{E}[\text{Tr}(z^T H_i z)] = \mathbb{E}[\text{Tr}(H_i z z^T)] = \text{Tr}(H_i \mathbb{E}[z z^T]) = \text{Tr}(H_i).$$

□

In practice the expectation is approximated by Monte Carlo:

$$\text{Tr}(H_i) \approx \frac{1}{m} \sum_{k=1}^m z_k^T H_i z_k.$$

Each Hessian-vector product $H_i z_k$ is computed without materialising H_i via a standard double-backpropagation identity: if $g_i = \nabla_{W_i} L$, then

$$\frac{\partial(g_i^T z)}{\partial W_i} = H_i z,$$

which is obtained by a second backward pass through the computation graph.

3. Solution Methods

3.1. HAWQ: Hessian-Aware Quantization

HAWQ [1] is a two-phase algorithm that uses the top Hessian eigenvalue to guide both the precision assignment and the sequential fine-tuning order.

3.1.1. Phase 1: Precision Assignment

Top eigenvalue via matrix-free power iteration. The dominant eigenvalue $\lambda_{\max}^{(i)}$ is computed by iterating:

1. Compute the gradient $g_i = \nabla_{W_i} L$ by a standard backward pass.

2. Initialise a random unit vector $v \leftarrow v/\|v\|_2$.
3. Repeat until convergence:
 - (a) Compute the scalar $s = g_i^T v$.
 - (b) Compute $Hv = \partial s / \partial W_i$ via a second backward pass.
 - (c) Update and normalise $v \leftarrow Hv / \|Hv\|_2$.
4. Extract the top eigenvalue via the Rayleigh quotient: $\lambda_{\max}^{(i)} = v^T H_i v$.

Sensitivity scoring and bit-width assignment. For each block i , compute $S_i = \lambda_{\max}^{(i)} / n_i$. Sort blocks by S_i in descending order. Assign conservative bit-widths (e.g., 8-bit) to high- S_i blocks and aggressive bit-widths (e.g., 2-bit) to low- S_i blocks, selecting from the hardware-supported precision set. Activation bit-widths are kept one step higher than weight bit-widths to accommodate high-variance dynamic signals.

3.1.2. Phase 2: Block-wise Fine-Tuning

Once bit-widths are locked, compute the shock metric $\Omega_i = \lambda_{\max}^{(i)} \|\Delta W_i\|_2^2$ for each block and sort in descending order. Execute quantization and fine-tuning strictly from highest to lowest Ω_i : the most structurally damaging block is absorbed first while the remaining blocks are still in full precision, giving the optimizer maximum flexibility to recover the loss landscape. After each block is tuned, its weights are frozen before proceeding to the next block.

Algorithm 1 HAWQ: Precision Assignment and Sequential Fine-Tuning

Require: Pre-trained blocks $\{W_i\}_{i=1}^b$, bit-width set $\mathcal{B}$, calibration data

- 1: // **Phase 1: Sensitivity scoring**
 - 2: **for** $i = 1$ to b **do**
 - 3: Compute $\lambda_{\max}^{(i)}$ via matrix-free power iteration
 - 4: $S_i \leftarrow \lambda_{\max}^{(i)} / n_i$
 - 5: **end for**
 - 6: Sort blocks by S_i descending; assign bit-widths from $\mathcal{B}$
 - 7: // **Phase 2: Block-wise fine-tuning**
 - 8: **for** $i = 1$ to b **do**
 - 9: Compute $\Delta W_i = Q_{B_i}(W_i) - W_i$
 - 10: $\Omega_i \leftarrow \lambda_{\max}^{(i)} \|\Delta W_i\|_2^2$
 - 11: **end for**
 - 12: Sort blocks by Ω_i descending
 - 13: **for** each block i in sorted order **do**
 - 14: Quantize W_i to B_i bits; fine-tune with all other frozen blocks
 - 15: Freeze W_i
 - 16: **end for**
-

3.2. HAWQ-V2: Trace-Weighted Quantization

HAWQ-V2 [2] addresses three limitations of HAWQ: (i) the top eigenvalue ignores the rest of the spectrum; (ii) bit-width selection is manual; (iii) activations are excluded from the second-order analysis.

3.2.1. Trace Estimation via Hutchinson’s Method

Replace $\lambda_{\max}^{(i)}$ with the normalized Hessian trace estimated stochastically:

$$\frac{1}{n_i} \text{Tr}(H_i) \approx \frac{1}{m n_i} \sum_{k=1}^m z_k^T H_i z_k,$$

where each product $H_i z_k$ is computed by double backpropagation as described in Section 2.4.

3.2.2. Mixed-Precision Activation Quantization

Unlike weights, activations $a_j(x_i)$ depend on the input x_i . For the empirical loss $L = \frac{1}{N} \sum_{i=1}^N \ell(x_i; \theta)$, the activation Hessian is block-diagonal across data points:

$$H_{a_j} = \text{diag}(H_{a_j(x_1)}, \dots, H_{a_j(x_N)}),$$

because $\frac{\partial^2 L}{\partial a_j(x_m) \partial a_j(x_n)} = 0$ for $m \neq n$. Consequently, trace estimation decomposes per sample and can be averaged:

$$z^T H_{a_j} z = \frac{1}{N} \sum_{i=1}^N z_i^T H_{a_j(x_i)} z_i,$$

making mixed-precision activation quantization tractable.

3.2.3. Automatic Bit-Width Selection via Pareto Frontier

HAWQ-V2 defines a global perturbation metric combining curvature with quantization distortion:

$$\Omega = \sum_{i=1}^L \text{Tr}(H_i) \|Q_{B_i}(W_i) - W_i\|_2^2.$$

Bit-width allocation is then cast as a constrained optimisation problem:

$$\min_{\{B_i\} \in \mathcal{B}} \Omega(\{B_i\}) \quad \text{s.t. model size} \leq C,$$

solved by enumerating the Pareto frontier of accuracy versus model size and selecting the configuration with minimal Ω . This removes the manual precision selection step required by HAWQ.

Algorithm 2 HAWQ-V2: Trace-Weighted Mixed-Precision Allocation

Require: Pre-trained blocks $\{W_i\}_{i=1}^b$, bit-width set $\mathcal{B}$, memory budget C , m Hutchinson samples

```

1: for  $i = 1$  to  $b$  do
2:   for  $k = 1$  to  $m$  do
3:     Sample  $z_k \sim \text{Rademacher}(n_i)$ 
4:     Compute  $H_i z_k$  via double backpropagation
5:      $\tau_k \leftarrow z_k^T H_i z_k$ 
6:   end for
7:    $\widehat{\text{Tr}}(H_i) \leftarrow \frac{1}{m} \sum_k \tau_k$ 
8: end for
9: for each admissible configuration  $\{B_i\} \in \mathcal{B}$  on the Pareto frontier do
10:   Compute  $\Omega = \sum_i \widehat{\text{Tr}}(H_i) \|Q_{B_i}(W_i) - W_i\|_2^2$ 
11: end for
12: return configuration  $\{B_i^*\}$  minimising  $\Omega$  subject to size  $\leq C$ 

```

3.3. Future Steps with PTQ4ViT

PTQ4ViT [3] addresses a complementary but distinct problem: given a fixed, uniform bit-width, how should the per-layer quantization *scaling factor* Δ be chosen? It uses the Hessian, that is approximated via the diagonal Fisher Information Matrix to define a weighted MSE objective for scaling factor search. This differs fundamentally from HAWQ-style methods, which use the Hessian to decide *which layers receive which bit-width*. Because our project focuses on the bit-width allocation question, PTQ4ViT falls outside our core experimental scope. However, the derivation chain from Taylor expansion through the Jacobian-Hessian relationship to the diagonal Fisher approximation provides valuable theoretical context. Moreover, at very low bit-widths (W4A4), PTQ4ViT’s own results show that mixed-precision allocation outperforms any uniform-precision baseline regardless of how accurately the scaling factor is tuned. It directly motivates the HAWQ/HAWQ-V2 approach we pursue. We identify combining both contributions (HAWQ-style allocation + PTQ4ViT-style calibration) as a natural direction for future work.

3.4. Implementation Overview

Our implementation is openly developed at <https://github.com/Fenix125/Hessian-Based-Mixed-Precision>. The proposed methods were implemented in PyTorch as a modular pipeline covering sensitivity estimation, bit-width allocation, and quantization-aware fine-tuning.

For HAWQ, layer sensitivity was estimated using a matrix-free power iteration procedure that computes the dominant eigenvalue of the Hessian via repeated Hessian-vector products obtained through double backpropagation. The resulting eigenvalues were normalised by the number of parameters to produce layer-wise sensitivity scores. These scores were further used to derive a fine-tuning schedule based on the magnitude of the expected loss perturbation.

For HAWQ-V2, sensitivity was estimated using Hutchinson’s stochastic trace estimator. Multiple random Rademacher vectors were sampled to approximate the trace of the Hessian through repeated Hessian-vector products. The estimates were averaged across several batches to reduce variance, enabling a stable approximation of layer-wise curvature.

Based on the computed sensitivity scores, layers were ranked and assigned bit-widths from a predefined set of precision levels. Higher precision was allocated to layers with higher sensitivity, while more aggressive compression was applied to less sensitive components. To ensure stability, certain layers directly connected to the output were kept at higher precision.

Quantization was implemented using symmetric uniform quantization with a straight-through estimator to preserve gradient flow during training. After assigning bit-widths, the model was optionally fine-tuned in a staged manner, where layers were progressively quantized and optimized according to their estimated impact on the loss.

4. Experiments and Results

Throughout both experiments we report top-1 accuracy, effective model size in MB (weight-only, computed from per-layer bit assignments), and accuracy drop relative to the FP32 baseline. Where multiple bit-width budgets are compared we additionally examine the Pareto frontier of accuracy versus model size.

4.1. Experiment 1. HAWQ: Staged Quantization-Aware Fine-Tuning

Setup. This experiment uses DeiT-small fine-tuned on the full Tiny-ImageNet training set (100,000 images, 200 classes) for one epoch, achieving a FP32 top-1 accuracy of 12.17% on the validation set. Hessian sensitivity was computed over the full training loader (batch size 32) on an NVIDIA RTX 5070 Ti. Candidate bit-widths were $\{8, 6, 4, 2\}$; the allocator produced an average of 5.04 bits per weight ($6.4\times$ compression). Four pipelines were compared: FP32 baseline, uniform PTQ W4A4, uniform QAT W4A4, and HAWQ mixed-precision with staged QAT (1 epoch, AdamW $\text{lr} = 10^{-5}$).

Figure 1 shows the first experiment, which isolated the effect of the HAWQ Hessian-guided fine-tuning schedule.

The sensitivity analysis revealed a span of over four orders of magnitude in S_i across the 72 blocks. Early attention layers dominate: block 0’s value-projection has $\lambda_{\max} = 232.7$ and its key-projection $\lambda_{\max} = 31.4$, while layers in blocks 5–11 cluster between 10^{-2} and 2.0. MLP blocks are consistently less sensitive per parameter than attention projections, because their larger parameter counts normalise away the curvature in $S_i = \lambda_i/n_i$. This heterogeneity drove the allocator to assign 8-bit precision to the sensitive early layers and 2–4-bit to flat mid-network blocks.

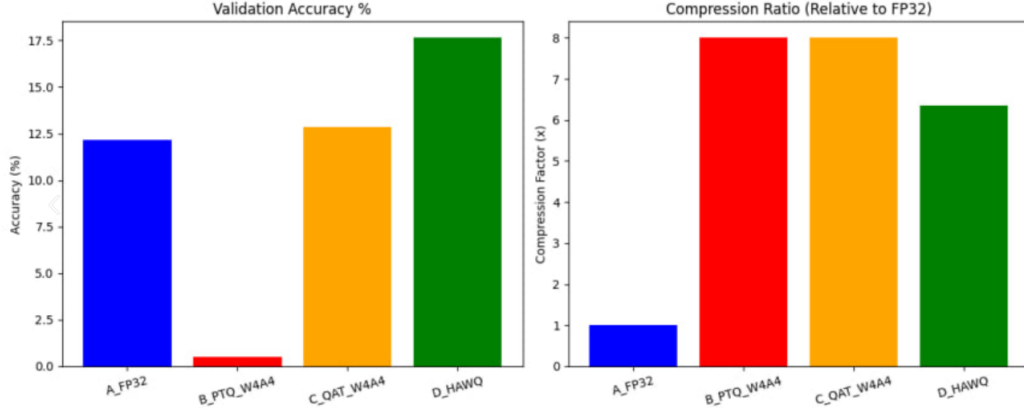


Figure 1: Comparison of FP32 baseline (A), uniform PTQ W4A4 (B), uniform QAT W4A4 (C), and HAWQ mixed-precision with staged QAT (D). Left: validation accuracy. Right: compression factor relative to FP32.

The Ω_i -ordered fine-tuning schedule produced a clear recovery trajectory over 73 sequential phases: validation accuracy rose from the post-quantization floor, with the highest-impact block (layer 3 MLP, $\Omega_i \approx 922$) absorbed first while all remaining weights were still in FP32, providing the optimizer maximal flexibility. The final HAWQ model reached **17.64%** top-1 accuracy against 12.17% for the FP32 reference in this shorter-training experiment, outperforming both uniform PTQ W4A4 (0.5%) and uniform QAT W4A4 (12.85%) at a $6.4\times$ compression ratio compared to $8\times$ for uniform 4-bit. This confirms that the deterministic fine-tuning order is itself a meaningful contribution: the same 1-epoch QAT budget yields substantially better recovery when applied in the Hessian-guided sequence.

4.2. Experiment 2. HAWQ-v1 vs. HAWQ-v2 Comparison

Setup. This experiment uses a stronger DeiT-small checkpoint fine-tuned for 5 epochs on 30,000 Tiny-ImageNet training images with AdamW and cosine scheduling, reaching 54.90% FP32 top-1 accuracy on 5,000 validation images. For Hessian analysis a small-batch loader (batch size 8) was used to keep second-order autograd graphs within GPU memory; both analyzers processed all 72 linear blocks in under 46 seconds. Candidate bit-widths were $\mathcal{B} = \{8, 6, 5, 4\}$; QAT recovery ran for 1 epoch with AdamW at $\text{lr} = 10^{-5}$. Seven pipelines were evaluated across four average bit-width budgets $\{4.5, 5.0, 5.5, 6.0\}$.

Table 1 and Figures 2 and 3 present the full comparison across seven pipelines and four bit-width budgets.

At 5-bit all three PTQ variants produce identical accuracy (49.68%), because the rank-based allocator at this budget assigns bit-widths that largely coincide with uniform 5-bit for most layers. The advantage of the Hessian-guided allocation is clearest at 4.5-bit: HAWQ-v1 and HAWQ-v2 achieve 30.86% and 30.48% respectively, compared to 4.80% for uniform PTQ, implying a gap of over 25 percentage points, confirming H2 at aggressive

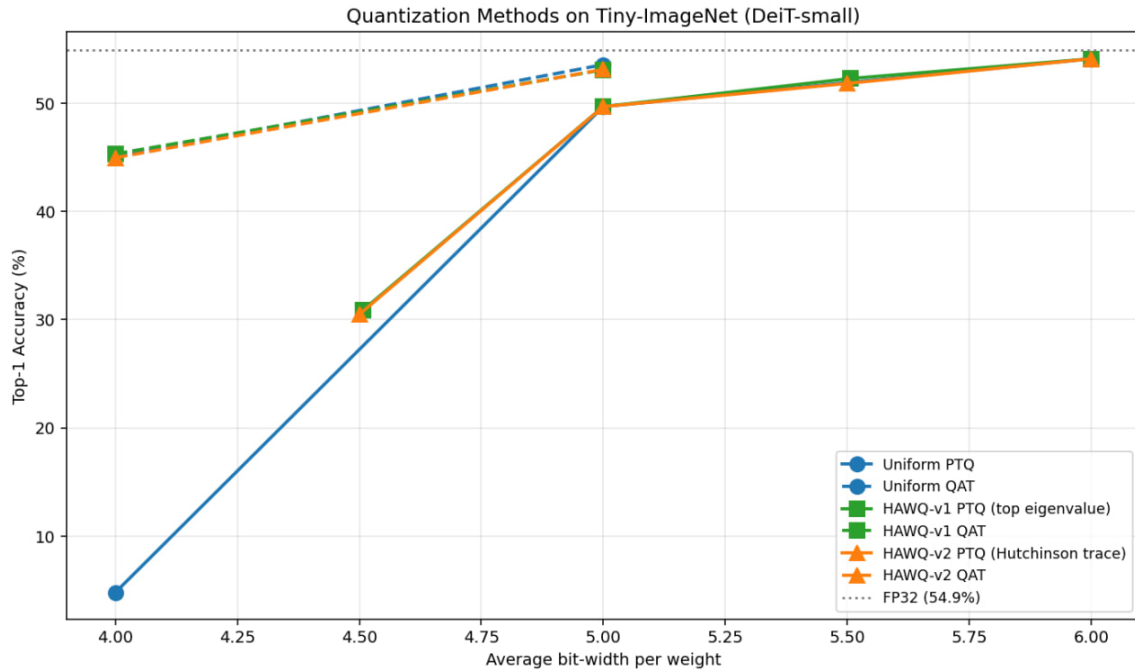


Figure 2: Top-1 accuracy vs. average bit-width per weight. Solid lines: PTQ. Dashed lines: QAT. The FP32 ceiling is shown as a dotted line at 54.9%. HAWQ-v1 and HAWQ-v2 track each other closely across all budgets; uniform PTQ collapses sharply at 4-bit while all Hessian-guided and QAT variants recover substantially.

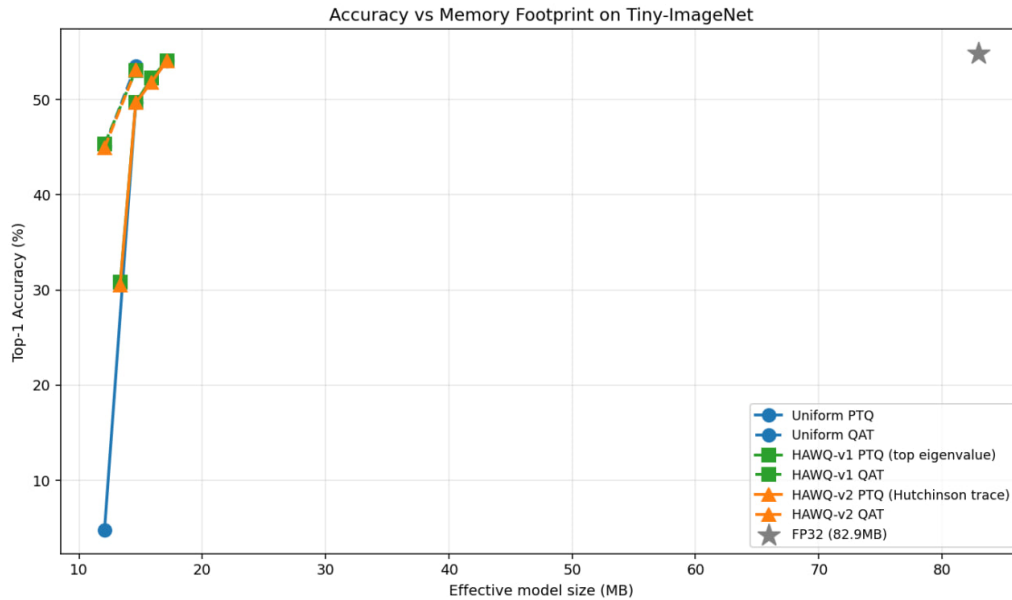


Figure 3: Pareto frontier of accuracy vs. effective model size. All quantized methods cluster between 12 MB and 17 MB. Hessian-guided PTQ holds a small advantage over uniform PTQ at 4.5-bit, while QAT recovery closes most of the gap to FP32 for all methods at 14.6 MB.

Table 1: Results at 5-bit average weight precision on Tiny-ImageNet (DeiT-small). Accuracy drop relative to FP32 baseline (54.90%).

Method	Avg bits	Size (MB)	Top-1 Acc. (%)	Δ vs FP32 (%)
FP32 baseline	32	82.9	54.90	0.00
Uniform PTQ	5	14.6	49.68	-5.22
HAWQ-v1 PTQ	5	14.6	49.68	-5.22
HAWQ-v2 PTQ	5	14.6	49.68	-5.22
Uniform QAT	5	14.6	53.56	-1.34
HAWQ-v1 QAT	5	14.6	53.08	-1.82
HAWQ-v2 QAT	5	14.6	53.12	-1.78

compression budgets where the loss from uniform quantization is catastrophic. Under QAT at 4-bit, all methods converge to similar accuracy ($\approx 45\%$), suggesting the 1-epoch recovery budget equalises performance once fine-tuning is applied.

The Spearman rank correlation between HAWQ-v1 and HAWQ-v2 sensitivity scores is $\rho = 0.865$, indicating strong but imperfect agreement. Divergence is concentrated in deep key-projection layers with very small eigenvalues but broader spectra, which is exactly the regime the trace ordering lemma predicts the two criteria should differ. However, because both methods use the same rank-based allocator, this ranking difference does not translate into a significant accuracy gap at the budgets tested (HAWQ-v2 QAT leads HAWQ-v1 QAT by 0.04 percentage points at 5-bit). A finer allocator exploiting the full trace information, such as the Pareto-based Ω optimisation of the original HAWQ-V2 paper, would provide a more discriminating test of H3.

5. Summary and Conclusions

5.1. Summary

This project approached model compression as a problem in applied numerical analysis. Starting from the second-order Taylor expansion of the task loss, we derived a rigorous quantization sensitivity metric, the normalized Hessian trace, and implemented both HAWQ (top-eigenvalue criterion) and HAWQ-V2 (Hutchinson trace criterion) for DeiT-small on Tiny-ImageNet. The key theoretical contributions formalised and implemented are: the quadratic loss approximation $\Delta L_i \approx \frac{1}{2} \Delta W_i^T H_i \Delta W_i$; the trace ordering lemma establishing $\frac{1}{n_i} \text{Tr}(H_i)$ as a more representative sensitivity criterion than $\lambda_{\max}^{(i)}$; Hutchinson’s identity enabling scalable matrix-free trace estimation; and the HAWQ shock metric $\Omega_i = \lambda_i \|\Delta W_i\|_2^2$ that determines the optimal sequential fine-tuning order.

5.2. Conclusions from Results

Our experiments allow us to evaluate the three project hypotheses directly.

H1: Sensitivity ordering supported. The per-layer sensitivity analysis reveals a span of over four orders of magnitude in S_i across the 72 linear blocks of DeiT-small. Layers with the highest scores (early attention value and key projections, $S_i \sim 10^{-3}$) are assigned 8-bit precision, while flat MLP blocks in deeper layers ($S_i \sim 10^{-6}$) receive 4-bit precision. The deterministic fine-tuning order generated from Ω_i scores is validated by the staged QAT experiment: processing the most disruptive blocks first (blocks 3 and 10, with $\Omega_i \sim 900$ and 40 respectively) before the numerically stable ones enables consistent validation accuracy recovery across phases.

H2: Mixed-precision benefit partially supported. In the PTQ regime at 5-bit average, HAWQ-v1 and HAWQ-v2 match uniform PTQ exactly (49.68%). This occurs because at this operating point the rank-based allocator produces an allocation that coincides with uniform 5-bit for most layers, and the bit-budget constraint allows only marginal differentiation. The advantage of mixed precision becomes visible at 4.5 bits: both HAWQ variants achieve approximately 30.5–30.9% accuracy versus 4.8% for uniform PTQ, a gap of over 26 percentage points. This confirms that the benefit of Hessian-guided allocation is most pronounced at aggressive compression budgets where the loss from uniform quantization is catastrophic. Under QAT recovery, HAWQ-v2 achieves 53.12% at 5-bit, 0.22 percentage points ahead of HAWQ-v1 (53.08%) and 0.44 points behind uniform QAT (53.56%), suggesting that the Hessian-guided allocation alone does not consistently dominate uniform allocation once QAT recovery is applied.

H3: Trace vs. eigenvalue partially supported. The Spearman rank correlation of $\rho = 0.865$ between HAWQ-v1 and HAWQ-v2 sensitivity scores indicates that the two criteria largely agree on layer ordering. The divergence is concentrated in deeper key-projection layers with broad but low eigenspectra, precisely the theoretical regime where the trace is expected to be a more informative criterion than $\lambda_{\max}$. However, because both methods use the same rank-based allocator and the same QAT recovery budget, the practical accuracy difference at the 5-bit operating point is small (< 0.1 percentage points). A more discriminating test would require a tighter budget (4-bit) or a finer-grained allocator that exploits the richer information in the trace.

5.3. Limitations

1. **Dataset and model scale.** All experiments used Tiny-ImageNet (200 classes, 64×64) rather than full ImageNet-1k. Sensitivity rankings and the relative benefit of mixed-precision allocation may differ on larger-scale settings where the loss landscape is higher-dimensional.
2. **Rank-based allocator coarseness.** Both HAWQ-v1 and HAWQ-v2 used a simple bucket allocator that partitions the sensitivity ranking into equal-size bins. At a 5-bit average budget with only four candidate bit-widths, most layers receive the

same assignment as uniform 5-bit, masking the potential advantage of the more informative trace criterion. A Pareto-based optimisation of $\Omega = \sum_i \text{Tr}(H_i) \|\Delta W_i\|_2^2$ as proposed in the original HAWQ-V2 paper would provide a more faithful test of H3.

3. **Limited QAT budget.** QAT recovery was constrained to 1 epoch to keep wall-clock time manageable. The staged QAT single-phase experiment demonstrates that the HAWQ fine-tuning schedule is beneficial in this budget; additional epochs would likely widen the gap between HAWQ-guided and uniform QAT.
4. **Hutchinson estimator variance.** With $m = 8$ samples and 2 batches, the trace estimates have non-negligible variance. Layers with near-zero curvature may receive noisy sensitivity scores that affect the allocation at aggressive budgets.
5. **Activation quantization.** Our experiments quantize weights only. Extending second-order sensitivity analysis to activations (as HAWQ-V2 proposes theoretically) remains unimplemented.

5.4. Future Work

1. **Pareto-based automatic bit selection.** Implementing the HAWQ-V2 constrained optimisation $\min_{\{B_i\} \in \mathcal{B}} \sum_i \text{Tr}(H_i) \|\Delta W_i\|_2^2$ over the full Pareto frontier would replace the coarse rank allocator and provide a fairer test of H3.
2. **Integration with PTQ4ViT scaling factor calibration.** Combining HAWQ-V2's bit allocation with PTQ4ViT's Hessian-guided scaling factor search would address both complementary decisions, especially how many bits each layer receives and how the quantization grid is positioned within those bits, in a unified pipeline.
3. **Variance reduction for Hutchinson estimation.** Replacing vanilla Rademacher sampling with variance-reduced estimators (Hutch++, XTrace) could maintain estimation quality with fewer samples, reducing the 42-second calibration cost and enabling more aggressive budget sweeps.
4. **Activation quantization.** Extending the second-order analysis to activations using the block-diagonal decomposition described in Section 3.2 would enable fully mixed-precision models and is a natural next step toward W4A4 deployment.
5. **Attention-head granularity.** The current allocator treats all linear projections within a transformer block as independent. Grouping the four projections of a single attention head and assigning a shared precision could improve allocation coherence, since query, key, and value projections participate jointly in the attention computation.

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